

L5. Time Evolution of Quantum Systems

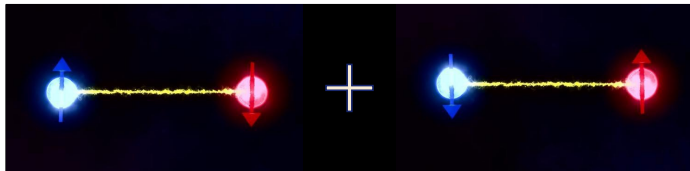
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Q Weirdness III: Entanglement in EPR Paradox

- In Bohm's version of the EPR paradox (local realism), the two particles are in the Bell state

$$|\psi_+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle).$$



- *Does color exist when no one is watching?*

- According to John Bell 30 years after the Bohr-Einstein debate, Alice and Bob can use the correlation (or anticorrelation) between their measurements to construct statistical quantities, which can be used to distinguish quantum theory from local hidden variable theories (in which spin values are determined by hidden variables not known to Alice and Bob, say, chosen by a virtual Charlie before the spin separation).
- As experiments have shown, Nature is deeply nonlocal, but the relativity theory is not violated by the nonlocality of quantum theory.

Outline

- Principle of Time Evolution
- Manipulating a Qubit: Rabi Oscillations
- Evolution of Density Matrices

5.1 Principle of Time Evolution

Take-Home Messages

- The unitary evolution of a state-vector with time is $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ with $U^\dagger(t)U(t) = I$.
- The evolution of the state obeys Schroedinger's equation

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = H|\psi(t)\rangle.$$

- The Hamiltonian H for a spin in a magnetic field $\vec{B}$ is

$$H = -\vec{\sigma} \cdot \vec{B}.$$

- For a time-independent magnetic field $\vec{B}$, the state-vector precesses about $\vec{B}$ on the Bloch sphere with an angular frequency $\omega_0 \propto B$.

Time-Evolution Operator

- Unlike in measurements, we do apply an operator on a state-vector when the system evolves in time.
- Given the state of a system $|\psi(0)\rangle$ at time $t = 0$, the state of the system at time t can be expressed as

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle,$$

where $U(t)$ is called the time-evolution operator.

- While measurements are probabilistic and irreversible in general, the time evolution of the state-vector is *deterministic and reversible*.

- We now have a Principle 5: The evolution of state-vectors with time is unitary, i.e.,

$$U^\dagger U = I.$$

- If U is unitary, and if $|a\rangle$ and $|b\rangle$ are any two state-vectors, then the inner product of $|a'\rangle = U|a\rangle$ and $|b'\rangle = U|b\rangle$ is the same as the inner product of $|a\rangle$ and $|b\rangle$.

- Proof:

$$\langle a'|b'\rangle = \langle a|U^\dagger U|b\rangle = \langle a|b\rangle.$$

- If $|a\rangle$ is normalized, so is $U|a\rangle$.
 - If $|a\rangle$ and $|b\rangle$ are distinguishable, i.e., $\langle a|b\rangle = 0$, so are $U|a\rangle$ and $U|b\rangle$. That is, information (or distinction) is never lost.

- We consider the time-evolution operator $U(\epsilon)$ for infinitesimal ϵ , and seek a differential equation for the evolution of the state-vector. The operator must satisfy

- Unitarity:

$$U^\dagger(\epsilon)U(\epsilon) = I.$$

- Continuity:

$$U(\epsilon) = U(0) + \frac{\epsilon}{i\hbar}H,$$

where $U(0) = I$ and $-i/\hbar$ is included by convention.

- Putting two things together, we have

$$\left(I + \frac{\epsilon}{i\hbar}H\right)^\dagger \left(I + \frac{\epsilon}{i\hbar}H\right) = I,$$

which, in the leading order $O(\epsilon)$, leads to $H^\dagger - H = 0$.

- Thus, we find that H defined in the continuity condition is Hermitian, $H^+ = H$.
- Further, H can be identified as the quantum Hamiltonian.
 - According to the definition, we can write

$$|\psi(t + \epsilon)\rangle = |\psi(t)\rangle + \frac{\epsilon}{i\hbar} H |\psi(t)\rangle.$$

- Taking the limit as $\epsilon \rightarrow 0$, we obtain

$$\lim_{\epsilon \rightarrow 0} \frac{|\psi(t + \epsilon)\rangle - |\psi(t)\rangle}{\epsilon/(i\hbar)} = i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = H |\psi(t)\rangle,$$

which is nothing but the time-dependent Schroedinger equation.

Example: Spin in a Magnetic Field

- If we know the Hamiltonian of a system (and its initial state), we can calculate how the state of the system evolves with time.
- But how do we get the Hamiltonian, e.g., for a single spin in a magnetic field?
 - H can be identified with the classical concept of a Hamiltonian, and its eigenvalues with energy.
 - In general, we can derive it from experiment, or borrow it from the classical theory.
 - Sometimes, we can pick one based on symmetry or just experience.

- In the case of a single spin, the choice is obvious.
- All 2×2 Hermitian matrix can be expanded as a linear combination of the identity matrix I and Pauli matrices.
 - Identity matrix means a constant energy. Nothing but the overall phase factor changes.
 - Higher powers of Pauli matrices cannot appear, as $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = 1$ and $\sigma_x \sigma_y = -\sigma_y \sigma_x = i\sigma_z$, etc.
- Spin is a 3-vector, and it must find another 3-vector to make a scalar operator H . In a magnetic field, the choice is

$$H \sim -\vec{\sigma} \cdot \vec{B} = - \begin{pmatrix} B_z & B_x - iB_y \\ B_x + iB_y & -B_z \end{pmatrix}.$$

- In particular, we consider a simple example in which $\vec{B} = B\hat{z}$. For convenience, we often write

$$H_0 = -\frac{\hbar\omega_0}{2}\sigma_z = \begin{pmatrix} -\frac{\hbar\omega_0}{2} & 0 \\ 0 & \frac{\hbar\omega_0}{2} \end{pmatrix},$$

in which we absorb B into the frequency ω_0 .

- The Hamiltonian has two eigenvalues $\pm\hbar\omega_0/2$, and the corresponding eigenstates are $|0\rangle$ and $|1\rangle$.

$$H_0 |0\rangle = -\frac{\hbar\omega_0}{2} |0\rangle, H_0 |1\rangle = \frac{\hbar\omega_0}{2} |1\rangle.$$

- The minus sign is a convention, such that $|0\rangle$ or “spin up” is the ground state.

- For a generic state $|\psi(t)\rangle = \alpha(t)|0\rangle + \beta(t)|1\rangle$, the time-evolution equation becomes, explicitly,

$$\begin{pmatrix} i\hbar \frac{\partial |\alpha(t)\rangle}{\partial t} \\ i\hbar \frac{\partial |\beta(t)\rangle}{\partial t} \end{pmatrix} = i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = H_0 |\psi(t)\rangle = \begin{pmatrix} -\frac{\hbar\omega_0}{2} \alpha(t) \\ \frac{\hbar\omega_0}{2} \beta(t) \end{pmatrix}.$$

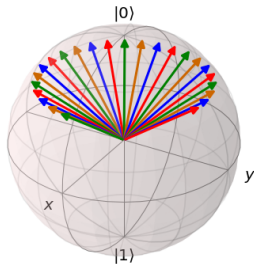
- Therefore, $\alpha(t)$ and $\beta(t)$ are decoupled, and we have

$$\alpha(t) = \alpha_0 \exp\left(\frac{i\omega_0 t}{2}\right), \quad \beta(t) = \beta_0 \exp\left(-\frac{i\omega_0 t}{2}\right),$$

where α_0 and β_0 can be determined by the state vector $|\psi(t=0)\rangle$.

- The initial state can be written, in the Bloch-sphere representation, as

$$|\psi(0)\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} |1\rangle.$$



- Therefore, the state-vector at time t is

$$|\psi(t)\rangle = e^{i\omega_0 t/2} \left[\cos \frac{\theta}{2} |0\rangle + e^{-i\omega_0 t} \sin \frac{\theta}{2} |1\rangle \right].$$

- The state-vector on the Bloch sphere precesses about $\hat{z}$ -axis with an angular frequency ω_0 .

- It is interesting to note that if we absorb the phase into a rotating basis

$$|0\rangle \rightarrow |0'\rangle \equiv e^{i\omega_0 t/2}|0\rangle, \quad |1\rangle \rightarrow |1'\rangle \equiv e^{-i\omega_0 t/2}|1\rangle,$$

the state “remains” unchanged as

$$|\psi(t)\rangle = \cos \frac{\theta}{2} |0'\rangle + \sin \frac{\theta}{2} |1'\rangle.$$

- In other words, in the so-called **rotating reference frame**, the state vector remains unchanged, i.e., the effective Hamiltonian in the rotating frame is $H' = 0$.

5.2 Manipulating a Qubit: Rabi Oscillation

- A constant $\vec{B} = B\hat{z}$ does not transform a state, e.g., from $|0\rangle$ to an arbitrary linear superposition of $|0\rangle$ and $|1\rangle$.
- The transformation can be done by applying, in addition to B , a transverse magnetic field $\vec{B}_1(t)$ rotating in the x - y plane with an angular frequency ω :

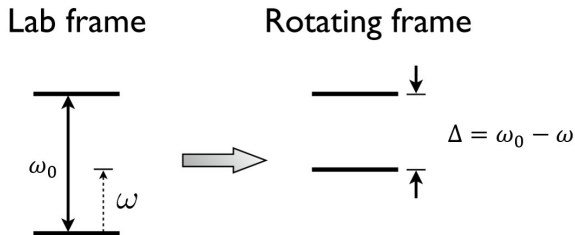
$$\vec{B}_1(t) = B_1(\cos\omega t\hat{x} - \sin\omega t\hat{y}),$$

such that the Hamiltonian becomes

$$H(t) = -\frac{\hbar\omega_0}{2}\sigma_z - \frac{\hbar\omega_1}{2}(\sigma_x \cos\omega t - \sigma_y \sin\omega t).$$

- The transverse field $\vec{B}_1(t)$ rotates in the x - y plane with the angular frequency ω . Let us introduce a reference frame that rotates with the field. In the rotating frame, the transverse field is now fixed (i.e., time-independent).
- In the absence of the transverse field, we have shown that the state vector precesses around the z axis with an angular frequency ω_0 . Now in the rotating frame, the state vector is expected to precess at a reduced angular frequency $\omega_0 - \omega$, which means the effective magnetic field along the z axis is decreased by a factor of $(\omega_0 - \omega)/\omega_0$.

- Pictorially, we have



- Accordingly, the effective Hamiltonian in the rotating frame becomes

$$H'(t) = -\frac{\hbar(\omega_0 - \omega)}{2}\sigma_z - \frac{\hbar\omega_1}{2}\sigma_x.$$

- This is identical to the Hamiltonian of a spin in a magnetic field along

$$\hat{n} = \left(\frac{\omega_1}{\Omega}, 0, \frac{\omega_0 - \omega}{\Omega} \right),$$

where $\Omega = \sqrt{(\omega - \omega_0)^2 + \omega_1^2}$. The magnitude of the magnetic field is proportional to $\hbar\Omega/2$.

- If at time $t = 0$ the spin is in the state $|0\rangle$, the spin is precessing about $\hat{n}$ with a characteristic frequency Ω .

- At a later time t , the probability of the system in state $|1\rangle$ can be calculated to be

$$|\langle 1|\psi(t)\rangle|^2 = |\langle 1'|\psi(t)\rangle|^2 = \left(\frac{\omega_1}{\Omega}\right)^2 \sin^2 \frac{\Omega t}{2}.$$

- This is the phenomenon of **Rabi oscillations**, which is the basic process to manipulate qubits in quantum computing. These oscillations are obtained by exposing qubits to periodic electromagnetic fields during suitably adjusted time intervals.

How to Calculate Time Evolution by Hand?

- In general, when H is independent of time, we can solve the time-evolution equation

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = H|\psi(t)\rangle$$

by symbolic integration that yields

$$|\psi(t)\rangle \equiv U(t)|\psi(0)\rangle = \exp\left[-\frac{iHt}{\hbar}\right] |\psi(0)\rangle,$$

where $|\psi(0)\rangle$ is the state-vector at $t = 0$.

- (Continued)

- In practice, we find all the eigenvalues and eigenvectors

$$H |j\rangle = E_j |j\rangle .$$

- Expand the initial state $|\psi_0\rangle$ with the eigenvector basis

$$|\psi(t=0)\rangle = \sum_j |j\rangle \langle j|\psi_0\rangle .$$

- The evolution of the state-vector is

$$|\psi(t)\rangle = \sum_j |j\rangle e^{-iE_j t/\hbar} \langle j|\psi_0\rangle .$$

5.3 Evolution of Density Matrices

- Quantum systems are inherently probabilistic. Random perturbations or measurements lead to probabilistic outcomes, which are described in terms of density matrices.
- For example, to prepare a quantum state $|\psi\rangle$, a noisy quantum system likely produces a random distribution of quantum states, i.e., $|\psi_i\rangle$ with probability p_i , due to imprecise controls or environmental noise. We use the density matrix

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

to describe the mixed state, as oppose to $\rho_0 = |\psi\rangle\langle\psi|$.

- The quantum operation formalism is a general tool for describing the evolution of quantum systems. Similar to how classical states evolves, quantum states transform as

$$\rho' = \mathcal{E}(\rho).$$

The map $\mathcal{E}$ in this equation is a **quantum operation**.

- The dynamics of a closed quantum system are described by a unitary transform. An unitary evolution is a quantum operation:

$$\mathcal{E}(\rho) = U\rho U^+,$$

which results from $|\psi'_i\rangle = U|\psi_i\rangle$.

The Liouville-von Neumann Equation

- For an infinitesimal evolution

$$U(\delta t) = e^{-i\mathcal{H}\delta t/\hbar} = 1 - \frac{i}{\hbar}\mathcal{H}\delta t + O(\delta t^2),$$

we expect

$$\begin{aligned}\rho(t + \delta t) &= \rho(t) - \frac{i\delta t}{\hbar} [\mathcal{H}\rho(t) - \rho(t)\mathcal{H}] + O(\delta t^2) \\ &= \rho(t) - \frac{i\delta t}{\hbar} [\mathcal{H}, \rho(t)] + O(\delta t^2),\end{aligned}$$

where we have used the commutator $[A, B] = AB - BA$.

- Taking the limit $\delta t \rightarrow 0$, we obtain

$$\dot{\rho}(t) = -\frac{i}{\hbar}[\mathcal{H}, \rho(t)] \equiv \mathcal{L}\rho(t),$$

where $\mathcal{L}$ is the so-called Liouvillian superoperator.

- The mixing rate of a state, i.e., the purity of a system, does not change due to the unitary Hamiltonian evolution, as

$$\frac{d}{dt}\text{Tr}\rho^2 = \text{Tr}\frac{d\rho^2}{dt} = \text{Tr}(2\rho\dot{\rho}) = \frac{1}{i\hbar}\text{Tr}\rho(\mathcal{H}\rho - \rho\mathcal{H}) = 0.$$

Measurement as Quantum Operation

- We can describe projective measurements in a similar way.
- For a pure state $|\psi\rangle$, consider an observable M , with a discrete spectrum $M = \sum_k \mu_k P_k$, where P_k denotes the projector corresponding to the k th eigenvector.
- The probability to obtain the measurement result μ_k is then given by

$$P(\mu_k) = \langle \psi | P_k | \psi \rangle = \text{Tr}(P_k |\psi\rangle\langle\psi|) = \text{Tr}(P_k \rho).$$

Obviously, this result can also be applied to a mixed state.

- The state of the system after measuring μ_k is given by

$$|\psi\rangle \longrightarrow \frac{P_k |\psi\rangle}{\sqrt{\langle\psi|P_k|\psi\rangle}}.$$

- In terms of density matrix ρ , the result is

$$\frac{P_k \rho P_k}{\text{Tr}(P_k \rho)}.$$

- The results have a more general formulation, in which the operators involved do not have to be projectors. Such a framework to describe typical quantum measurement is known as the positive operator-valued measurement.

- Therefore, we can identify projective measurement as a quantum operation as

$$\mathcal{E}(\rho) = \sum_k P_k \rho P_k,$$

if the result μ_k of the measurement is lost or unknown. Notice that the generic result is a mixed state, even for a pure initial state $\rho = |\psi\rangle\langle\psi|$.

- This may be used as the starting point for analysis of further operations on the system. Alternatively, such ignorance in our knowledge of the quantum state can be used to introduce the effect of quantum noise.

Summary

- The evolution of state-vectors with time is unitary. The infinitesimal evolution is generated by a Hermitian operator, known as the quantum Hamiltonian.
- A transversely rotating magnetic field leads to Rabi oscillations, which is the basic process to manipulate qubits in quantum computing.

References

- M. Le Bellac, A Short Introduction to Quantum Information and Quantum Computation (Cambridge, 2006), Chap. 3.
- L. Susskind and A. Friedman, Quantum Mechanics (Basic Books, 2014), Chaps. 8.
- J. J. Sakurai & J. Napolitano, Modern Quantum Mechanics, 2nd ed. (Addison-Wesley, 2011), Chap. 2.

Homework

P5-1. (a) Show that

$$\exp\left(i\frac{\theta}{2}\vec{\sigma} \cdot \hat{m}\right) = I \cos \frac{\theta}{2} + i(\vec{\sigma} \cdot \hat{m}) \sin \frac{\theta}{2},$$

where $\hat{m}$ is a unit vector and I the identity matrix. Hint: Calculate $(\vec{\sigma} \cdot \hat{m})^2$. The operator $\exp[i\theta(\vec{\sigma} \cdot \hat{m})/2]$ is the unitary operator $U[\mathcal{R}_{\hat{m}}(\theta)]$ which rotates by an angle θ about the $\hat{m}$ axis. (b) Choose $\hat{m} = (-\sin \phi, \cos \phi, 0)$, calculate $\exp[-i\theta(\vec{\sigma} \cdot \hat{m})/2]|0\rangle$ and $\exp[-i\theta(\vec{\sigma} \cdot \hat{m})/2]|1\rangle$. Compare your results with the solution to **P3-2**.

P5-2. In the lecture we showed that a spin placed in a magnetic field $\vec{B}(t) = B_0\hat{z} + B_1(\cos\omega t\hat{x} - \sin\omega t\hat{y})$ has a Hamiltonian

$$H(t) = -\frac{\hbar\omega_0}{2}\sigma_z - \frac{\hbar\omega_1}{2}(\sigma_x \cos\omega t - \sigma_y \sin\omega t).$$

Apply Schroedinger's equation, which is now a set of coupled differential equations, to show that if $|\psi(t=0)\rangle = |0\rangle$, then $|\psi(t)\rangle$ has a probability in state $|1\rangle$

$$P_{0\rightarrow 1}(t) \equiv |\langle 1|\psi(t)\rangle|^2 = \sin^2 \frac{\omega_1 t}{2},$$

in the case of resonance $\omega = \omega_0$.

P5-3. If A and B are non-commuting Hermitian operators, then $\exp[i(A + B)t]$ is a unitary operator. Show that

$$\exp[i(A + B)t] = \lim_{n \rightarrow \infty} \left[\exp\left(i\frac{At}{n}\right) \exp\left(i\frac{Bt}{n}\right) \right]^n.$$

We can use this expression for finite n as an approximation. A better approximation is

$$\exp[i(A + B)t] \approx \left[\exp\left(i\frac{Bt}{2n}\right) \exp\left(i\frac{At}{n}\right) \exp\left(i\frac{Bt}{2n}\right) \right]^n.$$